

Bimetallic Temperature Gauge (TG-BM)

The Bimetallic Temperature Gauge operates on the basis of coefficient of thermal expansion of metals. The bi-metal coil is manufactured from two cold-welded metal strips with different thermal expansion coefficients, which rotate in proportion to temperature. The rotary movement is conveyed to the pointer via a friction less assembly.

A comprehensive range of the standard version facilitates a variety of applications and uses. Adhering to the EN13190 standard, the design and construction of the bimetallic temperature gauge offers a robust and reliable means of temperature measurement with a quick response time.

The welded design system allows for best-in-class ingress protection (upto IP68 Standard). The use of corrosion resistant material, such as high grade stainless steel for both the casing and stem, makes this temperature gauge suitable to operate in an adverse environment and with corrosive media.



Available In

Nominal sizes	2 1/2"(63mm), 4"(100 mm) & 6"(150mm)
Ranges	0 to 50°C upto 400°C

Special Features

- Compliance to latest EN13190
- Standard ranges : 0 to 50°C upto 400°C
- Overload capacity 130% of FSD
- Ingress protection to IP68 4" & 6" models
- Accuracy class : tolerance class 1 as per EN13190
- Stainless steel casing
- External zero adjustment (optional)

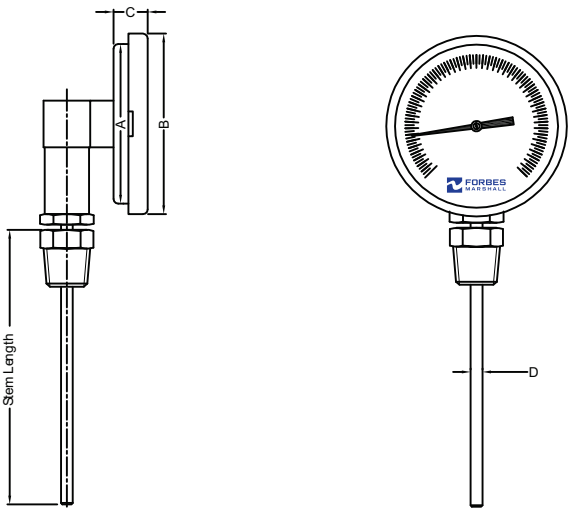
Industrial Applications

- Chemical / petrochemical industries
- Thermal, nuclear and hydroelectric power stations
- Pharmaceutical industries
- Food process industries / hygienic applications
- Water treatment plants
- Iron and steel industries
- Rubber moulding / processing plants
- Hydraulics and pneumatics
- Sugar industries / fertiliser industries
- Refineries
- Cryogenics

Standard Product Specifications	
Specifications	Description
Reference standard	EN 13190
Mounting	Direct bottom entry
Connection	½” NPT (M)
Case and Bezel	SS 304 (bayonet type bezel)
Stem diameter	6 mm
Stem	SS 316L
Fixed connection	SS 316
Sensing element	Bi-metal coil
Dial	Aluminum, white background with black marking
Pointer	Aluminum, black color micrometer with zero adjustable
Accuracy	Class 1 as Per EN 13190 (as standard)
Overload capacity	130% of maximum scale value
Window	Toughened glass
Protection	IP 68 – 4” & 6” models
	IP 65 – 2 1/2” model
	IP 65 – 4” compact model
Gasket	Silicon
Ambient temperature	-25°C to +65°C
Process temperature	Max. 400°C (for dry gauge)

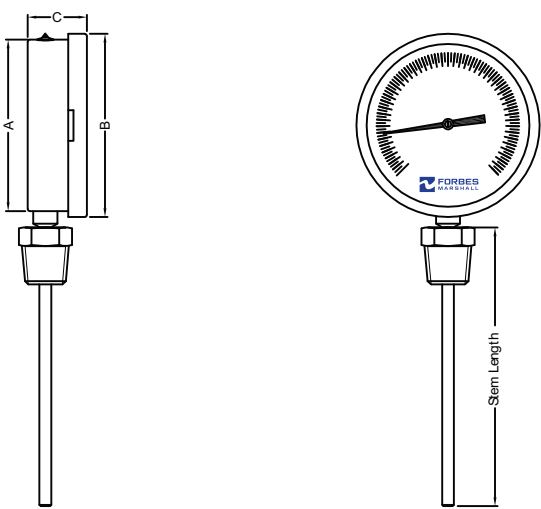
Mounting Options

Elbow Bottom



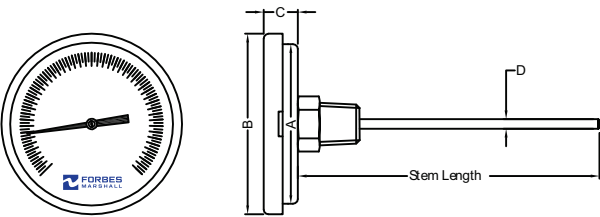
Parameters				
Dial Size	A	B	C	D
100	100	110	110	6/8/9.5/10
150	150	160	160	6/8/9.5/10

Direct Bottom



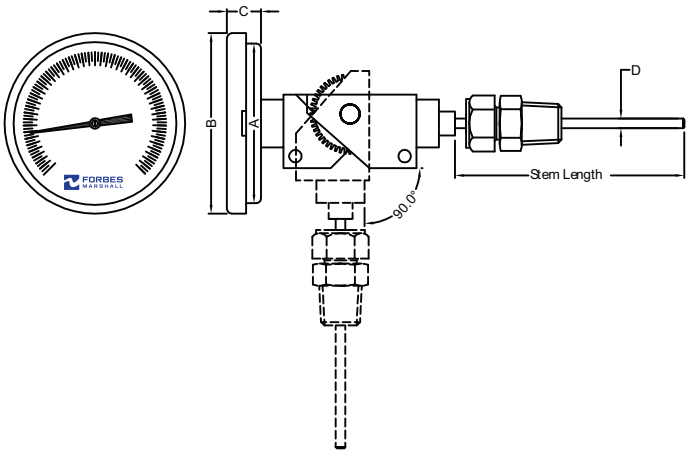
Parameters				
Dial Size	A	B	C	D
100	100	110	50	6/8/9.5/10
150	150	160	50	6/8/9.5/10

Center Back



Parameters				
Dial Size	A	B	C	D
100	100	110	21	6/8/9.5/10
150	150	160	24	6/8/9.5/10

Every Angle Mount



Parameters				
Dial Size	A	B	C	D
100	100	110	21	6/8/9.5/10
150	150	160	24	6/8/9.5/10

Standard Temperature Ranges				
Immersion Length vs Bulb Diameter				
Code	Temperature Range in Deg C	Minimum Immersion Length for Bulb Diameter 6 mm	Minimum Immersion Length for Bulb Diameter 8 mm	Minimum Immersion Length for Bulb Diameter 10 mm
A1	0 + 50	130	130	130
A2	0 + 60	120	120	120
A3	0 + 80	100	100	100
A4	0 + 100	90	90	90
A5	0 + 120	80	80	80
A6	0 + 150	70	70	70
A7	0 + 200	60	60	60
A8	0 + 250	60	60	60
A9	0 + 300	60	60	60
B0	0 + 400	50	50	50
E0	-20 + 40	120	120	120
E1	-20 + 60	100	100	100
E2	-20 + 120	70	70	70
E3	-30 + 30	120	120	120
E4	-30 + 50	100	100	100
E5	-30 + 70	90	90	90
E6	-40 + 40	100	100	100
E7	-40 + 60	90	90	90

General note: While selecting the range for your temperature gauge, please take into account the relative immersion length required for proper sensing. The maximum length recommended for bi-metallic temperature gauge should be taken as 700 mm.

Ordering Information

Model	
Bimetallic Adjustable connection	BA
Bimetallic Fixed connection (standard)	BF

Dial Size	
63 mm	D
100 mm	E
150 mm	F

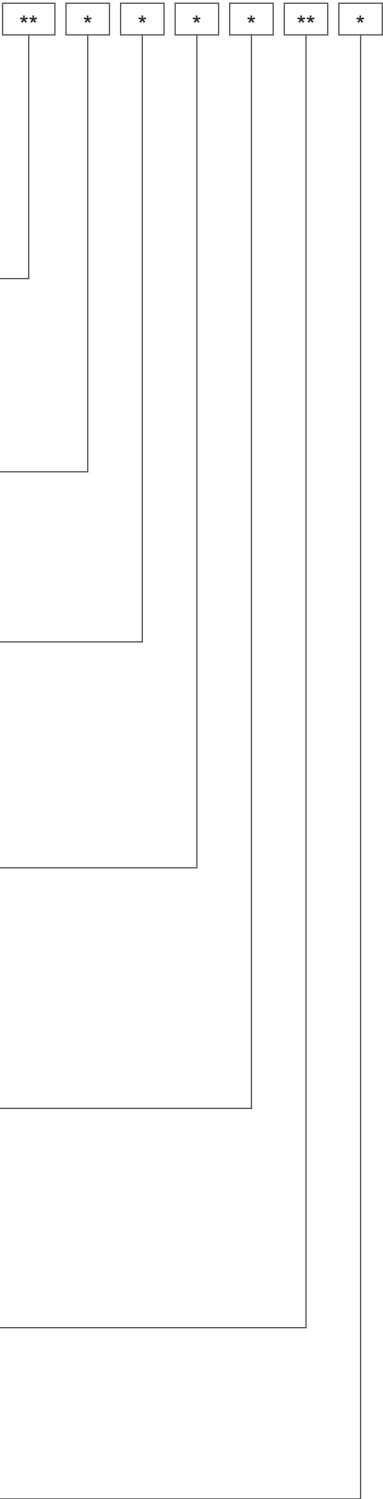
Mounting	
Every angle	0
Direct bottom	1
Center back	3

Connection Size	
1/4"	A
3/8"	B
1/2"	C
3/4"	D
M20x1.5P	1

Connection Type	
NPT(M)	A
BSP(M)	B
BSPT(M)	C
Metric thread (M)	M

Range	
Refer to standard temperature range table	

Engineering Units	
Degree celsius	A
Dual scale celsius / fahrenheit	C



Configuration Generation Code

Stem Diameter	Window
6 mm (standard)	Toughened glass (standard)
8 mm	Safety laminated shatterproof glass
10 mm	
Stem Length	Optional Features
300mm (standard)	110% of full scale reading (standard)
100mmTo 700mm with steps of 25mm - please specify	Dampening liquid silicon oil (upto 200°C)
	External zero adjustment
	Mirror scale dial for 6" dial size only
	Tag marking on dial
	SS tag plate
	Epoxy painted case and bezel
	Custom designed dial
	Special requirement - please specify
Case and Bezel	Tests and Certificates
SS 304 (standard)	Five point calibration certificate (standard)
SS 316	NABL traceability certificate
	Material test certificate
	EN 10204 3.1.B certificate
	Third party inspection
	Special requirement – please specify
Stem	
SS 316	
SS 316L (standard)	
Connection	
SS 316 (standard)	
SS 316L	
Pointer	
Fixed Pointer (standard)	

Ordering example: BAE1CAA5A-0000 (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

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